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A Barremian-Aptian (Early Cretaceous) North American paleomagnetic reference pole

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Abstract. Four intrusive complexes from southern Maine, containing rocks ranging in composition from olivine gabbro to granodiorite, were sampled to determine an Early Cretaceous reference pole for the North American plate. Sampling included the Lebanon diorite, the Tatnic Complex, Alfred Complex, and the Cape Neddick Complex dated radiometrically by the $^{40}Ar/^{39}Ar$ or K-Ar method on biotite between 125 and 120 Ma. The majority of samples have unblocking temperatures between 550°C and 580°C and Curie temperatures between 550°C and 570°C, indicating that the dominant remanence carrier is titanomagnetite, which crystallized from the magma or was produced by deuteric oxidation alteration during post-magmatic cooling. Titanomagnetite occurs as discrete C1-C3 grains, as exsolution or oxidation-exsolution lamellae in the silicates, as a product of magmatic reactions forming magnetite-ilmenite symplectites with pyroxene or olivine, or as ultrafine dust. Hemo-ilmenite is present as a minor component in the Lebanon Diorite, and, in this case, the remanence direction carried by the hemo-ilmenite is indistinguishable from the titanomagnetite. Detailed petrographic observations and rockmagnetic experiments demonstrate that, despite the coarse average grain size of the oxides in these rocks, which might lead one to expect unstable magnetic remanence, the magnetic remanence in most samples is extremely stable and caused by single-domain or pseudo-single-domain behavior in magnetite containing fine oxidation-exsolution lamellae of ilmenite. The Tatnic and Alfred intrusions are of normal polarity (D = 333°, I = 62°, pole 70.5°N, 210.3°E, A_{95} = 3.7°, N= 24 sites); the Cape Neddick and Lebanon intrusions are of reversed polarity (D = 161°, I = -57°, pole 73.7°N, 185.7°E, A_{95} = 5.0°, N= 17 sites). Together these intrusions yield a mean paleomagnetic pole at 72.2°N, 198.9°E, A_{95} = 3.3°, N = 41 sites. The well-characterized reversed polarity recorded in the precisely dated Cape Neddick Complex may represent the best age for the magnetic polarity chron M0r, the last reversal before the onset of the Cretaceous normal superchron at the Barremian-Aptian stage boundary.

Introduction

A paleomagnetic investigation of Early Cretaceous igneous intrusive complexes from southern Maine (Figure 1) was carried out to expand the very limited Early Cretaceous database for North America and with a view to (1) establishing an Early Cretaceous reference pole for North America, (2) defining the nature and stability of the remanence carrier(s), and (3) examining the behavior of the magnetic field during this period of apparent low secular variation prior to the onset of the Cretaceous normal polarity superchron. The investigated complexes intrude Paleozoic basement of Silurian to Lower Devonian age. They form part of the New England-Quebec intraplate magmatic province of *McHone and Butler* [1984]. Southeast of this province in the Atlantic lie the New England Seamounts [*Duncan,* 1984], comprising a total of approximately 2000 km of Mesozoic igneous activity [*Foland and Allen,* 1992]. Both *Duncan* [1984] and *Foland and Allen* [1992] suggest that the magmatism was related to a stationary mantle plume that was overridden by the North American plate during Cretaceous time.

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Apatite fission track studies indicate that this area has not been exposed to temperatures above 110°C since Cretaceous time [*Zimmerman et al.,* 1975; *Miller and Duddy,* 1989]. Furthermore, results of recent paleomagnetic work on the Triassic Abbott and Agamenticus intrusions [*Wu and Van der Voo,* 1988] and the Early Jurassic dikes on Seward and Doyle Points [*McEnroe and Brown,* 1992] are generally consistent with other Late Triassic and Early Jurassic poles and argue against any significant Cretaceous or younger regional overprinting or tilting. Thus the paleomagnetic directions obtained from this study are likely to represent primary directions. The sampled intrusions were mapped in detail by *Hussey* [1962]. Three of the intrusions are composed of multiple pulses. These were sampled to examine possible tilting within the intrusive complexes and to ensure a time-averaged direction for the calculated poles. Radiometric ages from biotite and hornblende have recently been reported by *Hubacher and Foland* [1991]. The overall geochronology of the New England region has been reviewed by Zartman [1988].

Geology and Site Localities

Alfred Complex

The Alfred Complex (Figure 2) is described by *Hussey* [1962]. It intrudes the Lower Devonian (?) Rindgemere

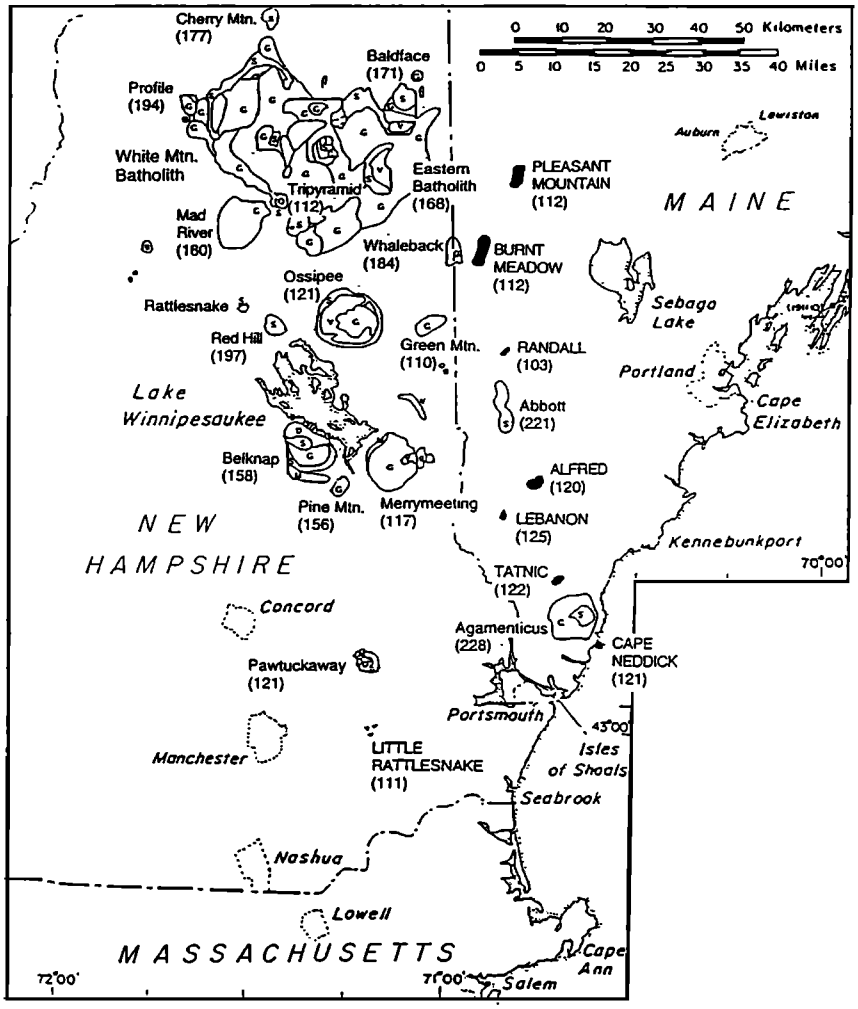


Figure 1. Map of southwestern Maine and adjacent New Hampshire and Massachusetts showing Mesozoic intrusions is shown (modified from *McHone and Butler* [1984]). Cretaceous intrusions in Maine including the Alfred, Tatnic, Cape Neddick, and Lebanon of this study are indicated in solid black.

Formation close to the northwestern contact of the Lyman pluton in the town of Alfred. The complex consists of four rock types in two major intrusions, an older cone-shaped intrusion 2.8 km in diameter of layered noritic gabbro with minor layers of anorthositic gabbro to the northeast; cut to the southwest by a circular complex 2.8 km in diameter consisting of outer monzodiorite and inner porphyritic granodiorite. *Hubacher and Foland* [1991] have obtained a $^{40}Ar/^{39}Ar$ age on biotite from the monzodiorite of the younger southwestern complex of 120.4±1.0 Ma, and *Foland and Faul* [1977] obtained a K-Ar age on biotite of 120±2 on the gabbro.

Fourteen sites were collected (Figure 2), nine from the older northeastern layered gabbro and five from the younger southwestern intrusion. Sites in the older intrusion include four in gabbros and two small mafic dikes on a traverse along a stream pavement on the middle branch of Mousam River in the southern part of the intrusion (1-6), two gabbros from the bed of the Mousam River in the northern part of the intrusion (10-11), and one gabbro from a hilltop at the extreme northern limit of the intrusion (9). The five sites from the younger intrusion include a monzodiorite and a monzodiorite with included gabbro from a highway exposure in the southeastern part (7,8), a monzodiorite from a railroad cut in the southeast (13), a granodiorite from a railroad cut in the south (14), and a granodiorite from a pavement outcrop to the north (12).

Tatnic Complex

The Tatnic Complex (Figure 3) is described by *Hussey* [1962]. It is intruded into the Silurian (?) Kittery Formation and Devonian quartz monzonite and granodiorite of the Webhannet pluton [*Fargo and Bothner*, 1995]. The complex consists of six rock types emplaced in four igneous phases. The oldest phase consists of volcanics that are preserved as xenoliths through much of the intrusion. The next phase is a cone-shaped complex, 3 km in diameter, of fine-grained gabbro grading upward into anorthositic gabbro. Volcanics and gabbros alike were then intruded to the southwest by an irregular area of olivine gabbro and poikilitic gabbro about 2.6 km in diameter, and this area was then further intruded by quartz diorite. *Foland and Faul* [1977] obtained a K-Ar age on biotite from the gabbro of 122±2 Ma and *Hubacher and Foland* [1991] obtained a $^{40}Ar/^{39}Ar$ age on biotite from the quartz diorite of 122±2 Ma.

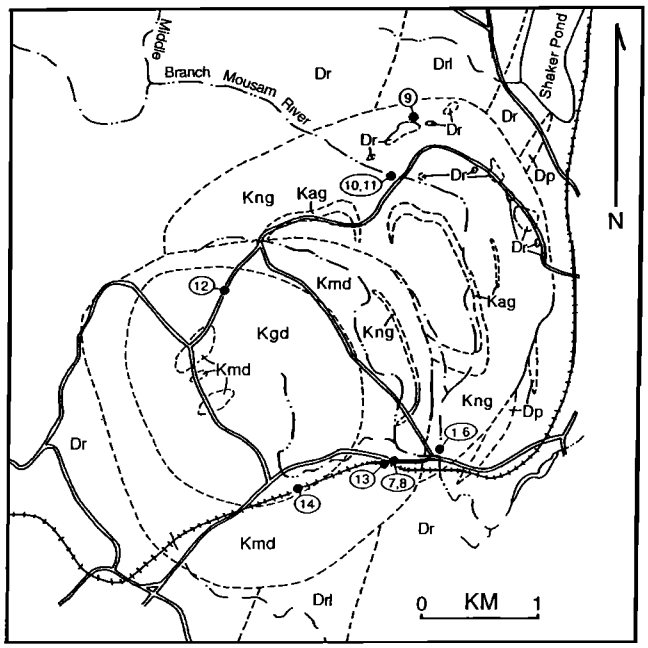


Figure 2. Geologic map of the Alfred Complex is shown (modified from *Hussey* [1962]) with locations of paleomagnetic sites. Geologic contacts, solid lines, accurately located; dashed lines, approximately located. Key to rock units: Devonian metamorphic and intrusive rocks; Dr, Rindgemere Formation; Drl, calc-silicate member of Rindgmere Formation; Dp, Devonian pegmatite. Members of the Cretaceous Alfred Intrusive Complex: Kng, noritic gabbro; Kag, anorthositic gabbro; Kmd, monzodiorite; Kgd, porphyritic granodiorite.

Of a total of 10 paleomagnetic sites (Figure 3), six were obtained in a quarry in the northern part of the gabbro, including four sites in gabbro and two from fine-grained mafic dikes intruding the gabbro (1-6). Of the four remaining sites, one is a coarse mafic porphyry assigned to the volcanics by *Hussey* [1962] (7), one is of the olivine gabbro (8), and two are from the youngest quartz diorite near the south edge of the intrusion (9, 10). *Hurley and Shearer* [1981] collected and demagnetized 18 paleomagnetic samples from the gabbroic part of the Tatnic Complex as part of their regional study on mid-Paleozoic rock units in New England.

Cape Neddick Complex

The Cape Neddick Complex (Figure 4) at York, Maine, is an oval intrusion of layered gabbro 1.26 x 0.83 km [*Hussey*, 1962]. It intrudes the low-grade, but vertically dipping Silurian-Ordovician Kittery Formation. Continuous coastal outcrops from the southeastern to the northern side of the intrusion provide the best exposures of any Mesozoic intrusion in New England. The complex involves five major rock types associated with two intrusive phases, plus a variety of mafic and felsic dikes. The earlier intrusive phase involves an intrusive agglomerate along some contacts, followed by normal gabbro that is chilled against it, and that grades inward to anorthositic gabbro. The later intrusive phase, confined to the interior of the complex, consists of olivine gabbro. This phase also grades inward toward higher plagioclase content. At the outer contact, the Kittery Formation is very strongly hornfelsed with development of cordierite-orthopyroxene assemblages in pelitic layers and wollastonite-diopside assemblages in calcareous layers. An $^{40}Ar/^{39}Ar$ age of 120.6±0.4 Ma on biotite from the gabbro is reported by *Hubacher and Foland* [1991].

The paleomagnetic localities are shown in Figure 4. All sites were collected as oriented blocks taken in inconspicuous locations so as not to mar the superb shore exposures. The sites include four of normal gabbro: three from near the northern contact (1,3,4) and one from the point opposite the Nubble (5); two of anorthositic gabbro from the northernmost shore exposures (6,7); two of cortlanditic gabbro, one from the eastern contact (8) and one from the western interior (11); two of dikes within the complex, one in normal gabbro near the northern contact (2) and one in anorthositic gabbro (9); two of dikes in country rock to the north of the complex (10, 14); and two of variably hornfelsed Kittery Formation (12, 13). Samples of the metamorphosed contact aureole and country rock were collected for a baked contact test.

Lebanon Diorite

The Lebanon Diorite is poorly exposed in an oval area (1.6 x 1.2 km) centered around Philbrick Road in the eastern part of Lebanon and the western part of Sanford, Maine [*Hussey*, 1962]. It intrudes the Rindgemere Formation of probable Devonian age and has essentially no topographic expression. A K-Ar age of 125 ± 3 on biotite Ma was reported for the intrusion by *Foland and Faul* [1977]. The diorite was redated by *Hubacher and Foland* [1991] and yielded a $^{40}Ar/^{39}Ar$ age of 132 Ma on biotite

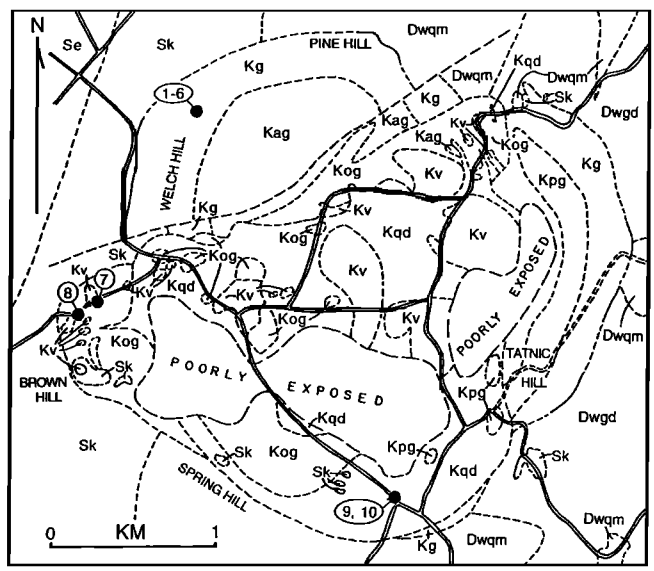


Figure 3. Geologic map of the Tatnic Complex is shown (modified from *Hussey* [1962]) with locations of paleomagnetic sites. Geologic contacts: solid lines, accurately located; dashed lines, approximately located. Key to rock units: metamorphosed Silurian stratified rocks; Sk, Kittery Formation; Se, Eliot Formation. Devonian intrusions of the Webhannet pluton; Dwgd, granodiorite; Dwqm, quartz monzonite. Cretaceous igneous rocks of the Tatnic Complex: Kv, volcanics undivided; Kg, gabbro; Kag, anorthositic gabbro; Kpg, poikilitic gabbro; Kog, olivine gabbro; Kqd, quartz diorite.

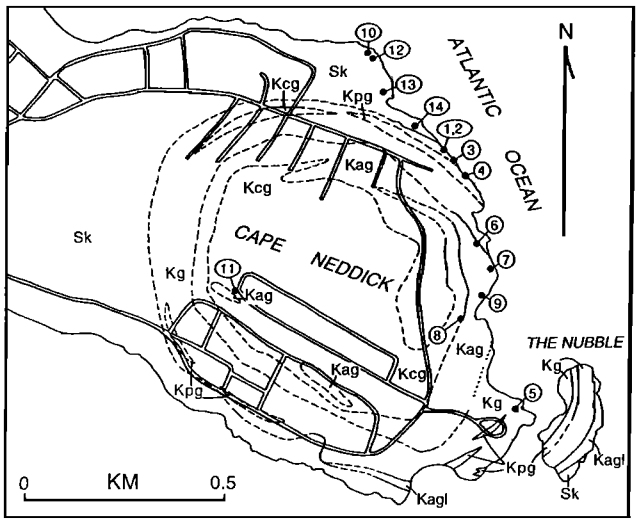


Figure 4. Geologic map of the Cape Neddick Complex is shown (modified from *Hussey* [1962]) showing locations of paleomagnetic sites. Geologic contacts: solid lines, accurately located; dotted lines, gradational contacts; dashed lines, approximately located. Key to rock units: metamorphosed Silurian stratified rocks; Sk, Kittery Formation. Members of the Cretaceous Cape Neddick Complex; Kagl, agglomerate; Kg, normal gabbro; Kag, anorthositic gabbro; Kpg, pegmatitic gabbro; Kcg, cortlanditic gabbro.

[*Hubacher and Foland*, 1991] but with a discordant spectrum. The age of 132 Ma is considered to be influenced by excess argon and the true age is probably closer to 125 Ma (K.A. Foland, personal communication 1991), within the error of the age cited by *Foland and Faul* [1977]. Therefore, the age of 125 ± 3 is used here for working purposes.

Three sites in fresh diorite were collected in a very low exposure on the southeast side of Philbrick Road. Two additional diorite sites were obtained from large rounded natural exposures about 50m southeast of Philbrick Road near the Lebanon-Sanford town line. Samples of the contact-metamorphosed Rindgemere Formation were collected for a baked contact test. Thin sections and polished cores from the five diorite samples show that it is composed of plagioclase and dusty K-feldspar, augite, red-brown biotite, titano-magnetite, hemo-ilmenite, ilmenite, leucoxene, and minor chlorite, green biotite, hematite, pyrite, and apatite.

Paleomagnetism

A common conception in paleomagnetism is that stable remanence is held in grains of single-domain and pseudo-single-domain size (<8 μm), and that grains of multidomain (MD) size, such as would be expected in coarse-grained intrusive igneous rocks would not retain a stable remanence over a geological timescale. Nevertheless, several workers [*Graham*, 1953; *Larson et al.*, 1969; *Strangeway et al.*, 1968; *Hargraves and Ade-Hall*, 1975] have suggested features of multidomain-sized grains that may cause them to behave like single-domain or pseudo-single-domain material in terms of magnetic stability. Because the present study involves mainly coarse-grained rocks, it has required both routine alternating-field and thermal demagnetization procedures, and special investigations using reflected-light microscopy and rock magnetic experiments to verify the nature and stability of the magnetic remanence results.

Field and Laboratory Procedure

Samples were collected in the field using a portable gasoline-powered drill and oriented with a Brunton compass, and whenever possible, a sun compass. Oriented blocks were collected from sites where it was difficult to drill or where long traverses were necessary.

The natural remanent magnetization (NRM) of the 237 samples were measured with a Molspin spinner magnetometer at the Paleomagnetics Laboratory at the University of Massachusetts, Amherst, or the Paleomagnetics Laboratory at the Norwegian Geological Survey (NGU) in Trondheim, Norway. At least two specimens from each site were selected for detailed alternating field (AF) and/or thermal demagnetization treatment. For AF treatment, specimens were progressively stepwise demagnetized at increments of 5 or 10 mT up to a peak induction of 100 mT using a three-axis tumbling (NGU) or a single-axis nontumbling Schoenstedt GSD-1 AF Geophysical specimen demagnetizer (UMASS). The alternating field demagnetization method was effective in isolating the characteristic remanence from secondary components in all samples. Remaining samples were AF-demagnetized in a minumum of three intervals. The fields selected were based on the detailed AF experiments.

For thermal demagnetization, specimens were heated in air in steps between 50° and 100°C to temperatures of 585°C or 630°C using a noninductively wound furnace with an accuracy of ± 5°C and cooled in a field-free chamber at NGU and measured on magnetometer in an area with an ambient field of < 5 nT. Remanent magnetization was determined after each step.

Hysteresis properties and Curie temperatures were measured on a vibrating sample magnetometer (Princeton Applied Research) at the Institute for Rock Magnetism, University of Minnesota. Hysteresis curves were measured at room temperature in a maximum field of 0.8T.

Data were plotted on standard orthogonal vector end point diagrams [*Zijderveld*, 1967]. The primary remanent magnetization direction was determined by Spectrum [*Smethurst*, 1987], or when necessary by Linefind [*Kent et al.*, 1983], or by standard paleomagnetic techniques. Site mean directions and the 95% confidence limits were calculated by standard Fisher statistics [*Fisher*, 1953]. Thirty-nine of the 41 sites yielded site mean directions with 95% confidence ($\alpha 95$) of less than 10°. All sites were used in calculating the Early Cretaceous paleomagnetic pole.

Remanence Behavior

The description of remanence behavior is divided into two parts based on magnetic polarity, the normal polarity Tatnic and Alfred Complexes and the reversed polarity Cape Neddick Complex and the Lebanon Diorite.

Normal polarity. The natural remanent magnetization intensities of these samples range from 0.2 to 13 A/m. The NRM directions showed some scatter. During alternating field (AF) demagnetization, samples showed the presence of a soft component which was removed by 20 mT. This component was either of normal or reversed polarity (Figure 5). At higher fields a single component with an univectorial decay heading toward or reaching the origin is present. Vectors regressing toward the origin were successfully isolated in each sample from these intrusions by either thermal or AF

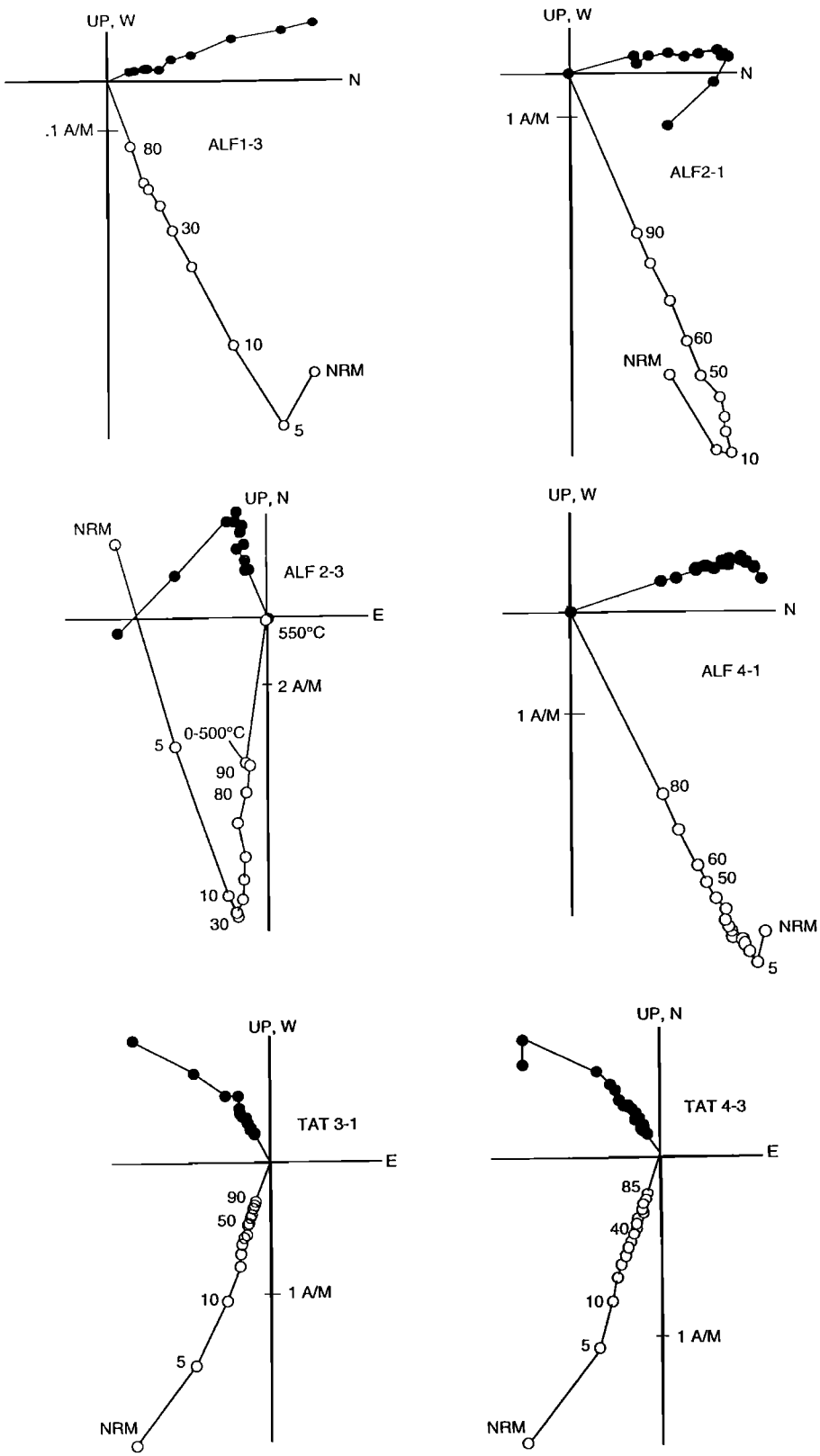


Figure 5. Vector end point diagrams are shown for samples from the Alfred and Tatnic intrusions.W, west; E, east; S, south; N, north. Also shown are solid circles, horizontal component; open circles, vertical component; numbers adjacent to circles, alternating fields in milliteslas or temperature in °C for thermal demagnetization. Also, origin is included in vector. All samples show a high coercivity component. Samples unblock between 500° and 580°C as shown in sample ALF 2-3.

demagnetization. The vector end point demagnetization diagrams shown in Figures 5 and 6 are typical of the specimens from these sites. The coercivity of the specimens varied from highest in the gabbros, with a median destructive field (MDF) from 40 to 90 mT, to lowest in the monzodiorites with an MDF from 0.5 to 10 mT. Several sites from the Alfred gabbro showed an unusually high coercive component with a MDF of 90 mT. To test for the presence of hematite, which has higher coercivity and thermal unblocking temperature, some samples were thermally demagnetized after AF demagnetization to 90 mT. Sample ALF 2-3 was subjected to incremental thermal demagnetization up to its unblocking temperature of 550°C (Figure 5). Though this sample has a high coercivity, the thermal demagnetization conclusively demonstrates that magnetite is the primary magnetic carrier. This has been verified in reflected light microscopy and by isothermal remanent magnetization (IRM) experiments. Although there are a variety of coercivities found in the samples (Figure 7a), little change in direction was evident above 20mT.

One sample from each site was demagnetized thermally. In general, the dominant unblocking temperature was between 500°C and 580°C (Figure 7b). These temperatures are indicative of titanium-poor magnetite with up to ~20% ulvöspinel in solid solution to pure end member magnetite as the primary remanence carrier. The thermal demagnetization and stepwise AF procedures produced similar, if not identical, results. Figure 6 shows the decay paths of two samples from the olivine gabbro (site 8) in the Tatnic intrusion. Virtually identical vectors were obtained above an unblocking temperature of 500°C and under AF demagnetization above 10 mT.

Reversed polarity. Most samples show a normal overprint, which is easily removed at low fields. After the removal of a low coercivity component, most samples show a single reversed component, with an univectorial decay heading toward or reaching the origin. During the AF demagnetization experiments, samples from the Lebanon Diorite show a harder coercive component, though the characteristic remanent direction is unblocked by 20 mT and a univectoral decay follows (Figure 6). More than 30% of the NRM intensity remains after peak fields of 90 mT. The samples from the Cape Neddick Complex have a range of coercivities with medium destructive fields between 10 mT and 90 mT, although the characteristic remanent directions were recovered after fields of 30 mT. The vector end point demagnetization diagrams shown in Figure 6 are typical of the specimens from these sites. AF demagnetization procedures were successful in isolating the characteristic remanent magnetization directions in most samples.

Unblocking temperatures up to 580°C, indicative of magnetite as the remanence carrier, are present in the majority of the Cape Neddick and Lebanon samples. In a few samples from the Lebanon Diorite a small component persists up to 630°C above the Curie temperature of magnetite at 580°C. This high-temperature component has the same characteristic remanent direction as the lower-temperature unblocked component (580°C). It is carried by hemo-ilmenite. The alternating field and thermal demagnetization procedures gave similar characteristic remanence directions for both the Lebanon Diorite and the Cape Neddick Complex (Figure 6).

Magnetic Mineralogy, Stability, and Relaxation

Introduction. The paleopoles obtained in this study are significantly steeper than those obtained in many previous paleomagnetic studies of Cretaceous rocks in North America. To guard against a misinterpretation due to a contamination from the present-day field that would steepen the inclination, it is necessary to examine mechanisms that could produce a present-day overprint that was not removed by the extensive demagnetization procedures. Two hundred and sixty-seven samples were subjected to careful stepwise alternating field or thermal demagnetization. In general, a soft secondary magnetization, that was easily removed at low fields or temperatures, was followed by a linear decay representing the stable component. No evidence for additional overprints were observed. A question that needs to be addressed is whether these rocks could have been overprinted by relaxation of magnetic domains, by a thermal process, or by a chemical process, in a manner which is difficult to detect. Rock magnetic measurements made are on bulk samples (rock chips, powders, or cores) and are not representative of any individual feature. By combining optical observations, demagnetization spectra of natural remanent magnetization (NRM), Curie temperatures, and demagnetization spectra of isothermal remanent magnetization (IRM) and anhysteritic remanent magnetization (ARM), an attempt was made to understand what mineral phase or phases is associated with each component of the remanence.

Petrographic observations related to demagnetization behavior. Sixty-four polished cores or thin sections representative of the paleomagnetic collection were examined to identify the magnetic carriers and to examine for alteration that could result in a change of the remanent magnetization. Observations of the opaque minerals, made in oil immersion in reflected light, included size, morphology, texture, degree of oxidation, and exsolution features. Optical observations were combined with measured rock magnetic properties and demagnetization behavior to assess the reliability of the signal and the timing of acquisition of magnetization.

When titanomagnetite grains are subjected to oxidation-exsolution of ilmenite, the remaining magnetite is more Ti-poor , and hence, has a higher Curie temperature than the original titano-magnetite. The oxidation-exsolution process may also produce a smaller effective magnetic grain size because the ilmenite lamellae parallel to the {111} planes of magnetite isolate the areas of Ti-poor titanomagnetite. The new compositions and the textures produced by oxidation-exsolution may have different and important rock magnetic properties as compared to the original titano-magnetite. The oxidation states of the observed titanomagnetites fall into classes C1 (unexsolved) to C3 (moderately exsolved) of *Haggerty* [1976]. Ilmenite intergrowths in the titanomagnetites are classified into three types: sandwich, trellis, and composite [*Buddington and Lindsley,* 1964]. The morphologies of the titanomagnetites range from skeletal to cruciform and herringbone, and to euhedral.

Fine-grained titanomagnetites are found in the numerous dikes which cut the intrusions. Commonly, these dikes contain equant and skeletal homogeneous C1 titanomagnetite grains, and C2 grains with trellis ilmenite lamellae and minor sandwich-type lamellae. The C1 grains are found in the chilled margins of the dikes. They progress from skeletal or herring bone to euhedral morphology. The paleomagnetic directions obtained from these dikes are very stable with a linear decay to the origin after the removal of a small overprint as is shown in the vector end point diagram (Figure 5) of a dike from the Tatnic gabbro (site 4). It is not surprising to have a high

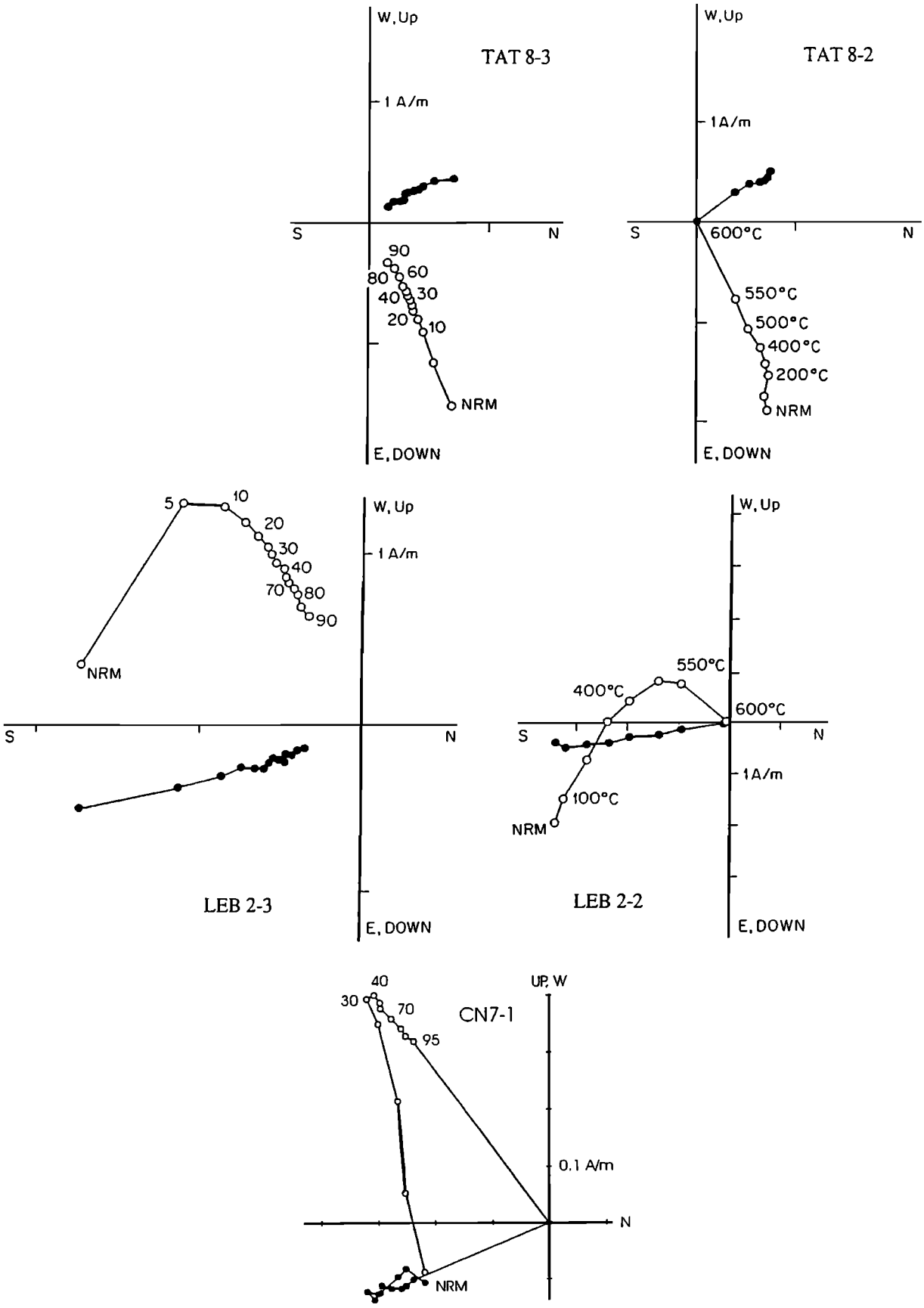


Figure 6. Comparison of the effects of alternating field (AF) and thermal demagnetization on paired samples from Tatnic and Lebanon intrusions is shown. For each pair, the two demagnetization methods yield similar final directions. AF demagnetization from the Cape Neddick intrusion (sample CN7-1) shows large normal overprint followed by a linear decay toward origin. See Figure 5 for further explanation.

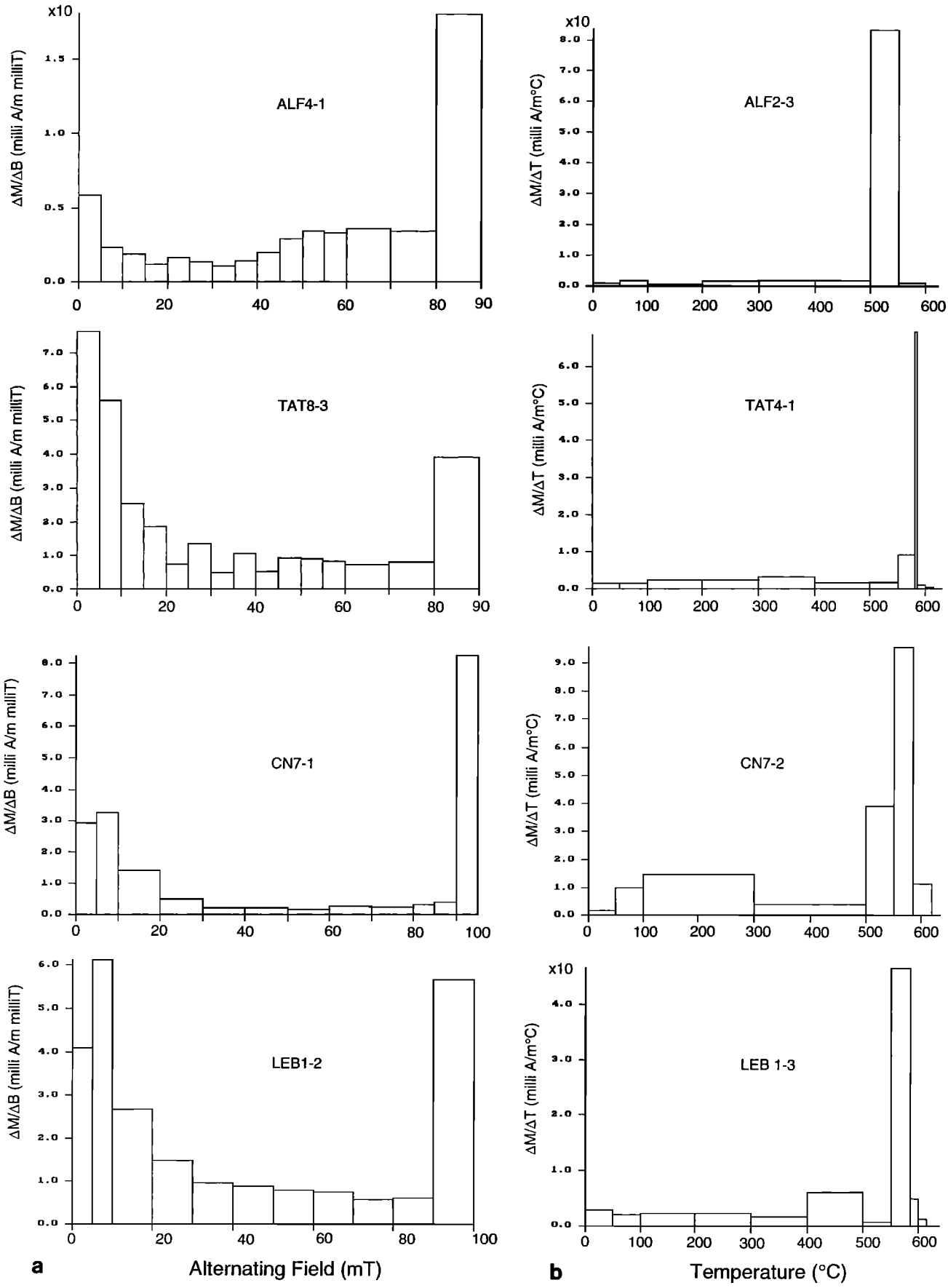


Figure 7. Magnetic stability diagrams is shown for representative samples from the four Cretaceous intrusions. (left) AF stability and ΔM/ΔB equals change in magnetic moment / change in applied field in milliamps per millitesla. Samples show significant amount of magnetization remaining after AF demagnetization to 90mT. (right) Thermal stability and ΔM/ΔT equals change in magnetic moment / change in temperature in milliamps per °C. Most samples unblock in the interval of 500° - 585°C clearly indicating magnetite as the predominant magnetic mineral.

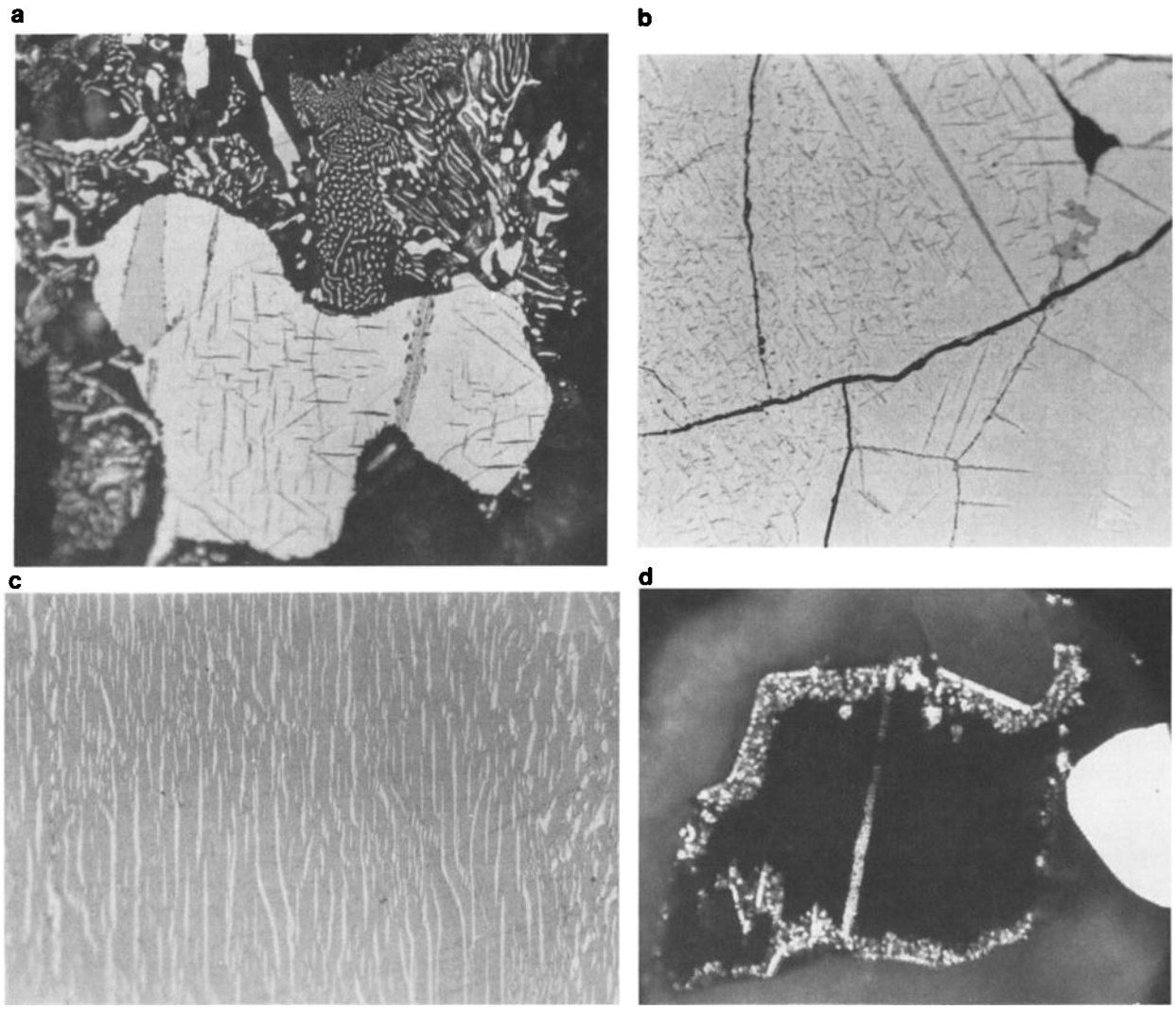


Figure 8. Reflected-light photomicrographs, all photo-graphed with a blue filter. a) Cape Neddick Gabbro. Lower center is a large titanomagnetite grain with {111} trellis type oxidation exsolution of ilmenite (gray) and sandwich lamellae with fine granular exsolution of pleonaste (black) adjacent to titanomagnetite. Surrounding the titanomagnetite is a symplectite of magnetite-ilmenite-pyroxene. Olivine is adjacent to the symplectite. Olivine - magnetite intergrowths are also present. These symplectites contain a very fine grained magnetite. Width of view 225μ. b) Cape Neddick Gabbro. Titanomagnetite grain with very-fine {111} trellis oxidation exsolution of ilmenite (gray) lamellae and with fine granular exsolution of pleonaste (black) at the ilmenite-titanomagnetite interfaces. The faint cloth-like texture [*Ramdohr*, 1969] in the centre of the photograph which is relatively free of ilmenite lamellae is composed of very, very fine magnetite (<< 0.5μ) "protoilmenite". This texture indicates original exsolution of ulvöspinel from magnetite which was later oxidized to ilmenite. Note the brighter areas around the larger ilmenite lamellae (~7μ long <0.5μ wide). These are magnetite-rich areas where the ilmenite has diffused to form lamellae. Width of view 50μ. c) Discrete ilmenite (gray) from the Lebanon Diorite with fine lamellae of hematite (white). Width of view 55μ. d) Lebanon Diorite. Fine-grained rim and intergrowth of magnetite and Ti-rich biotite. Width of view 200μ.

stability component associated with titano-magnetites of skeletal or herringbone morphology.

In the gabbros of the Cape Neddick Complex (Figure 8a) magnetite grains with oxidation-exsolution of ilmenite along {111} plane of magnetite are observed. The ilmenite lamellae are either sandwich or trellis type. Many samples show abundant, very fine trellis ilmenite lamellae that are less than 0.5 μ in diameter. Some titanomagnetites show exsolution of ulvöspinel and pleonaste (Mg-Fe-Al spinel) in addition to the oxidation-exsolution of ilmenite (Figure 8a,8b). After oxidation-exsolution of ilmenite from the titanomagnetite grains, very fine granular pleonaste exsolved from the magnetite along the interface with the ilmenite lamellae. The small trellis lamellae subdivide the larger titanomagnetite grain (>>100 μ) into very small discrete areas of magnetite. These small areas of magnetite may be highly anisotropic due to the number of sets of lamellae that are oxy-exsolved. At high magnifications (1200x) in oil, a very fine clothlike texture is observed (Figure 8b). These areas lack abundant ilmenite lamellae. This texture is due to exsolution of ulvöspinel from magnetite parallel to the {100} plane of the magnetite. The ulvöspinel was later oxidized to ilmenite,

termed "protoilmenite" by *Ramdohr* [1969]. Ilmenite, rather than ulvöspinel, is inferred due to the slight optical anisotropy that is observed. The two phases, present in the mottled clothlike texture, each smaller than half a micron, are the lighter areas of magnetite, and the darker regions of ilmenite. This large MDsize grain (>100μ) has undergone oxidation and exsolution, significantly modifying the original large grain, and creating textures that produce discrete magnetite areas that are in the pseudo-single-domain (PSD) or single-domain (SD) size range. Also shown in Figure 8a is a symplectite of very fine to fine-grained magnetite and ilmenite with pyroxene. Olivine is adjacent to the symplectite and in some cases appears to be involved with it. These symplectites are very fine grained, and there is a clear shape anisotropy in many of the symplectites. The symplectites also show oxidation-exsolution of magnetite and ilmenite at the micron scale in the PSD region. All these features in the Cape Neddick gabbro appear to reduce the effective grain size and increase coercivity.

In the Lebanon Diorite, a small high-temperature component which persists up to 620°C and has a high coercivity, is carried by hematite which has exsolved along {0001} planes in ilmenite (Figure 8c). Many of these lamellae are thinner than 0.5μ. These intergrowths, require a miscibility gap between these rhombohedral phases. Ilmenite-hematite forms a solid solution above the consolute point which has been shown to lie below 650°C and is asymmetric toward hematite [*Lindsley*, 1991]. Other contributors to the higher coercivity component in the Lebanon Diorite could be exsolved grains of magnetite in the silicates. A common observation is titanium-rich biotite with reaction rims of ilmenite and magnetite (Figure 8d).

Curie and unblocking temperatures in relation to mineralogy. Curie temperatures were measured on samples from the four intrusions. These were analyzed either as rock chips or powders. The samples showed a discrete range of Curie temperatures from 540°C to 570°C, indicating a Ti-poor titanomagnetite or pure magnetite as the principal magnetic mineral. Most samples produced good reversible heating and cooling curves indicating little oxidation or alteration of the Ti-poor titanomagnetite during the thermomagnetic runs. In the thermal stability diagrams (Figure 7b), plotted as vector differences, variations in thermal blocking temperatures are shown. Some samples, such as Tatnic 4-1 and Alfred 2a-3, show one discrete unblocking temperature, at 550°C and 585°C, respectively, while others indicate a slight stepped unblocking spectrum. Samples from the Lebanon intrusion (LEB 2-4) show stepped unblocking temperatures from 500°C to 620°C. Discrete titanomagnetite grains (C2) with trellis-type lamellae of ilmenite were observed in the Lebanon Diorite which would account for the unblocking temperatures up to 580°C and ilmenite grains with {0001} lamellae of hematite would account for the highest temperature portion of the spectrum.

Grain size and magnetic relaxation. Multidomain size grains were observed in all samples and, in fact, are the volumetrically dominant size magnetic particles. An interesting feature is that most of the stability diagrams from the mafic parts of the intrusions display a significant component of PSD- and/or SD- type behavior with a MDF greater than 30mT (Figure 7). As concluded from Curie and thermal un-blocking temperatures, and optical observations, this coercive component is carried by titanomagnetite. The majority of samples have MDFs above 20mT which is considered the maximum coercivity for MD magnetite [*Bailey and Dunlop*, 1983, *Dunlop*, 1983]. The questions that remain are how were the effective magnetic grains formed, exsolved, or modified, and when was the remanence blocked?

Titanomagnetites with {111} oxy-exsolution lamellae of ilmenite are the typical magnetic grains in the Tatnic, Alfred, and Cape Neddick intrusions. Grain sizes of titanomagnetite range from 0.5 mm down to the optical resolution of the microscope (about 0.4 μm). The majority of grains are significantly above the grain size limit for PSD grains ~8μ [*Moskowitz and Banerjee*, 1979]. Optically, on the basis of size, the majority of these grains would be classified as multidomain and therefore unstable carriers of remanent magnetization over the geological timescale of interest. This interpretation contradicts the bulk measurements showing that the field needed to remove the NRM ranges from 30mT to greater than 90mT. The question then arises as to whether some of the MD-sized grains (> 8μ) are behaving as PSD grains and are in fact the remanent magnetic carriers.

It may be that the very fine exsolution features (Figure 8) are creating areas of stable pseudo-single-domain grains within the multidomain-sized grains. Examples of such a fine-scale exsolution are found in the normal gabbro at Cape Neddick Complex, and in the Alfred and Tatnic intrusions, where exsolution of ulvöspinel and pleonaste (Mg-Fe-Al spinel) in titanomagnetites and oxidation-exsolution of ilmenite (Figure 8b) are abundant and on a fine scale. The very fine granular pleonaste exsolved from the magnetite along the interface with the ilmenite lamellae, are possibly producing regions that are highly stable magnetically. In the Alfred, Tatnic, and Cape Neddick intrusions graphic intergrowths of magnetite and titanium-rich biotite and symplectites of very fine to fine-grained magnetite and ilmenite with pyroxene have oxidation textures in the PSD size range. In addition, some olivines in the Cape Neddick sample are oxidized, and contain a very fine grained magnetite that would be in the region of PSD and a stable carrier of the thermal remanent magnetization (TRM).

NRM, ARM, and IRM demagnetization and magnetic mineralogy. All samples have unblocking temperatures in the magnetite range. In addition, a small contribution from hematite, up to 620°C, is found only in the Lebanon intrusion. Also indicative of magnetite as the primary magnetic mineral are IRM runs which show samples saturated by fields of 30mT (Figure 9), clearly indicating magnetite as the primary magnetic mineral. In Figure 10, comparative curves of the demagnetization of NRM, ARM, and IRM are shown. In all samples the NRM and the ARM are more resistant to the AF demagnetizing field than the IRM indicating a significant component of SD or PSD behavior [*Lowrie and Fuller*, 1971; *Bailey and Dunlop*, 1983]. Typical gabbro samples show a very high coercivity that indicates primarily PSD behavior by their resistance to AF demagnetization; 50% of the intensity (J/J_0) typically remains after 50-80mT. Most samples show a mixed behavior of PSD grains and MD grains.

All samples indicate some MD behavior as shown by the removal of components at low field (< 20mT) (Figure 7a), then display a high coercive component as shown by an increased resistance to the AF field. Many samples have 30 to > 50% of the NRM remaining after applied AF fields of 80mT. The *p*TRM remaining is commonly greater than 1 A/m and can be greater than 2 A/m. Because the *p*TRM is so high, it seems

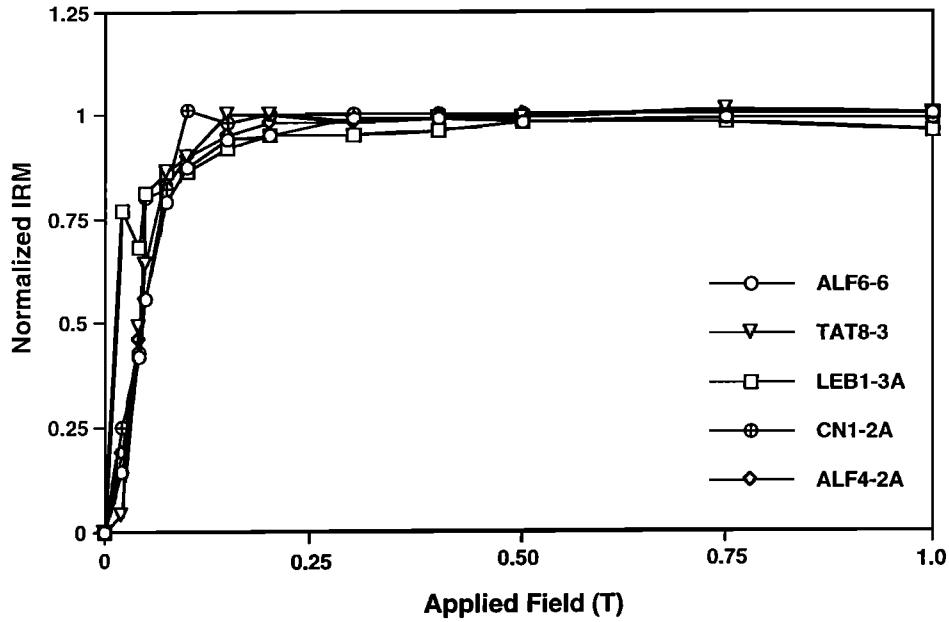


Figure 9. Isothermal remanent magnetization (IRM) acquisition diagram showing normalized IRM (magnetization applied normalized to saturation magnetization). Field is in Teslas. Almost all samples saturate by 0.2 T indicating titanomagnetite or magnetite as the primary magnetic mineral.

probable that some of the original homogeneous multidomain grains, that is those with very fine textures, were modified into aggregates of pseudo-single and single-domain particles by oxidation and/or exsolution. This concept has been suggested for years [*Graham*, 1953; *Larson et al.*, 1969; *Strangeway et al.*, 1968; *Hargraves and Ade-Hall*, 1975]. It has also been rejected by many authors [*Stacey and Banerjee*, 1974; *Hargraves and Banerjee*, 1973] in support of the concept that the stable NRM is carried by fine exsolution lamellae in the silicate grains. It appears that the silicate grains are playing only a minor role in the high-stability spectra of these samples. From a volume perspective, it is difficult to account for the total amount of magnetization remaining above fields >30mT, in which MD magnetite grains should not be contributing, without attributing some of the stability to grains in the MD size range. Thus it seems probable that some grains in the MD size range have been modified into aggregates of pseudo-single-domain grains by oxidation-exsolution processes during post-magmatic cooling.

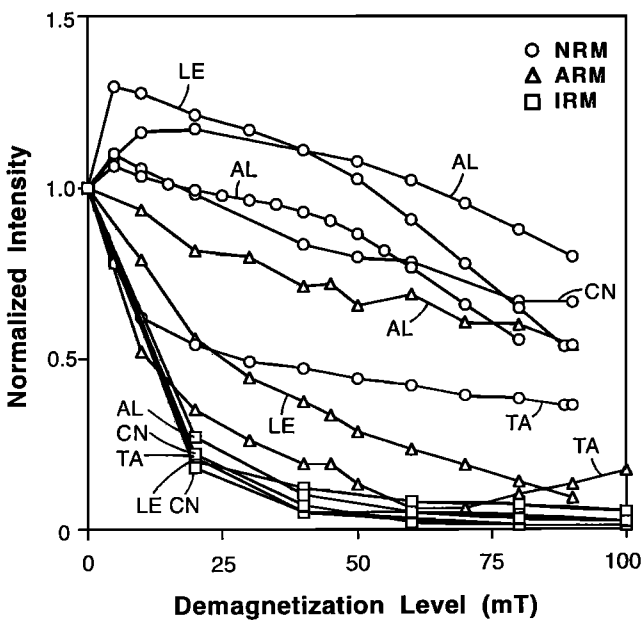


Figure 10. Demagnetization of NRM, ARM, and IRM normalized to demagnetization level is shown and field is in milliteslas. The IRM curves decay more rapidly than the ARM or NRM curves indicating PSD-behavior in the samples.

Paleomagnetic Results

Paleomagnetic data for the Alfred, Cape Neddick, Lebanon, and Tatnic intrusions are listed in Tables 1a and 1b. The data are divided into two sets based on magnetic polarity. The normal polarity set consists of 24 sites (149 samples) from the Tatnic and Alfred Complexes. The reversed polarity set consists of 18 sites from the Cape Neddick Complex and the Lebanon Diorite (88 samples). The site mean directions for the Alfred intrusions (N = 14 sites) and the Tatnic intrusions (N = 10 sites) are shown in Figure 11. In the Alfred and Tatnic intrusions there is excellent clustering of the data with no evidence of tilting after intrusion, either within or between the intrusions. The site mean directions for the Cape Neddick intrusions (N = 12 sites) and the Lebanon Diorite (N = 6 sites) are shown in Figure 11. These intrusions show good clustering of the data and little evidence of tilting within or between the intrusions after cooling.

The Cape Neddick Complex and the Lebanon Diorite pass a contact test. In addition to these positive contact tests, recent paleomagnetic work on the nearby Triassic Abbott (225 Ma) and Agamenticus (228± 5) plutons by *Wu and Van der Voo* [1988] and Early Jurassic dikes on Seward and Doyle Points [*McEnroe and Brown*, 1992] show little evidence of regional overprint, and the results are generally consistent with other Late Triassic and Early Jurassic poles. These Triassic and

Table 1a. Summary of Paleomagnetic Mean Data for the 125-120 Ma Southern Maine Intrusions

Site	N/N_o	D	I	α_{95}	k	Plat°N	Plong°E
			Alfred Complex				
Alfred 1	5/5	338	65	2.4	1438	74.0	221.9
Alfred 2	5/5	350	69	3.6	452	78.6	256.8
Alfred 3	5/5	337	69	5.3	206	72.1	238.3
Alfred 4	4/4	330	60	7.7	143	67.7	203.4
Alfred 5	7/7	338	63	1.8	1086	74.1	211.1
Alfred 6	8/8	324	64	7.1	116	64.4	217.3
Alfred 7	7/7	346	60	3.9	240	79.4	191.2
Alfred 8	6/6	336	58	5.5	149	71.4	193.1
Alfred 9	6/6	347	55	2.4	755	77.4	165.9
Alfred 10	6/6	352	55	4.8	374	80.1	150.5
Alfred 11	5/5	348	57	5.3	209	79.3	171.4
Alfred 12	7/7	314	60	1.9	1062	56.2	211.3
Alfred 13	7/7	338	68	5.4	128	73.2	234.6
Alfred 14	8/8	324	69	4.4	158	64.5	233.0
Mean	14/14	338	63	3.6	121	74.0	209.8
			Tatnic Complex				
Tatnic 1	6/6	304	61	2.2	891	49.6	217.6
Tatnic 2	5/5	323	60	2.7	807	62.6	207.3
Tatnic 3	6/6	327	61	2.3	877	65.8	208.0
Tatnic 4	7/7	327	60	2.0	870	65.5	205.3
Tatnic 5	6/6	324	69	3.0	490	64.4	233.2
Tatnic 6	8/8	323	61	2.8	379	63.0	209.7
Tatnic 7	6/6	343	58	9.3	53	79.9	176.9
Tatnic 8	10/10	322	59	1.8	792	61.6	205.5
Tatnic 9	5/5	325	68	4.8	257	65.2	230.0
Tatnic 10	4/4	342	50	3.7	630	71.1	164.3
Mean	10/10	327	61	4.6	113	65.7	208.4
			Cape Neddick Complex				
CN 1	8/8	181	-55	4.1	188	82.3	103.1
CN 2	5/6	179	-44	8.1	89	72.6	112.2
CN 3	4/4	174	-51	6.9	180	77.6	133.6
CN 4	4/4	166	-50	2.6	146	73.3	155.7
CN 5	5/5	153	-68	20.0	12	70.2	232.0
CN 6	3/3	157	-57	3.9	1521	72.1	192.4
CN 7	5/5	152	-50	4.0	363	64.6	179.3
CN 8	4/4	154	-57	5.7	265	69.5	192.2
CN 9	5/5	166	-53	3.7	433	75.4	162.5
CN 10	3/3	156	-56	12.1	105	71.0	190.2
CN 11	6/6	145	-58	7.2	141	63.4	201.5
CN 12	2/2	154	-59	8.7	821	70.3	198.2
Mean	12/12	162	-55	4.6	91	75.6	173.8
			Lebanon Diorite				
LEB 1	8/8	166	-62	3.1	319	79.8	204.1
LEB 2	6/7	164	-58	3.4	380	77.1	183.9
LEB 3	6/6	162	-54	1.4	2146	73.6	173.3
LEB 4	6/6	145	-62	3.1	458	64.7	211.4
LEB 5	7/7	136	-56	3.2	349	56.0	203.0
Mean	5/5	158	-59	7.5	113	73.3	194.3

N/N_o ratio of number of samples analyzed to number used in determining site mean directions; D, I, mean declination and mean inclination; α_{95} and *k Fisher* [1953] statistical parameters; Plat°, Plong°, latitude (N°) and longitude (°E) of magnetic pole.

Table 1b. Summary of Paleomagnetic Mean Data for 125-120 Ma. Southern Maine Intrusions

Intrusion	N/N_o	D	I	α_{95}	k	Plat°N	Plong°E	A_{95}
Tatnic Complex	10/10	327	61	4.6	113.2	65.7	208.4	-
Alfred Complex	14/14	338	63	3.6	121.4	74.0	209.8	-
Cape Neddick	11/12	162	-55	4.6	76.2	74.9	175.2	-
Lebanon Diorite	5/5	158	-59	7.5	104.8	73.2	195.1	-
Combined Reversed Mean	16/17	161	-57	-	84.5	73.7	185.7	5.0
Combined Normal Mean	24/24	333	62	-	111.7	70.5	210.3	3.7
Combined Mean	41/41	336.3	59.9	-	85.6	72.2*	198.9*	3.3
Selected Mean	40/41	336.4	59.7	-	85.4	72.3*	198.2*	3.3

N/N_o ratio of sites used in determining intrusive mean directions to the number of sites demagnetized; D, I, mean declination and mean inclination; α_{95} and *k Fisher* [1953] statistical parameters; Plat°, Plong°, latitude (N°) and longitude (°E) of magnetic pole. Reversed directions are inverted to upper hemisphere, average includes sites from Cape Neddick gabbro and Lebanon diorite. Normal average includes sites from Alfred and Tatnic intrusions. Plat* and Plong* are calculated from site level VGPs.

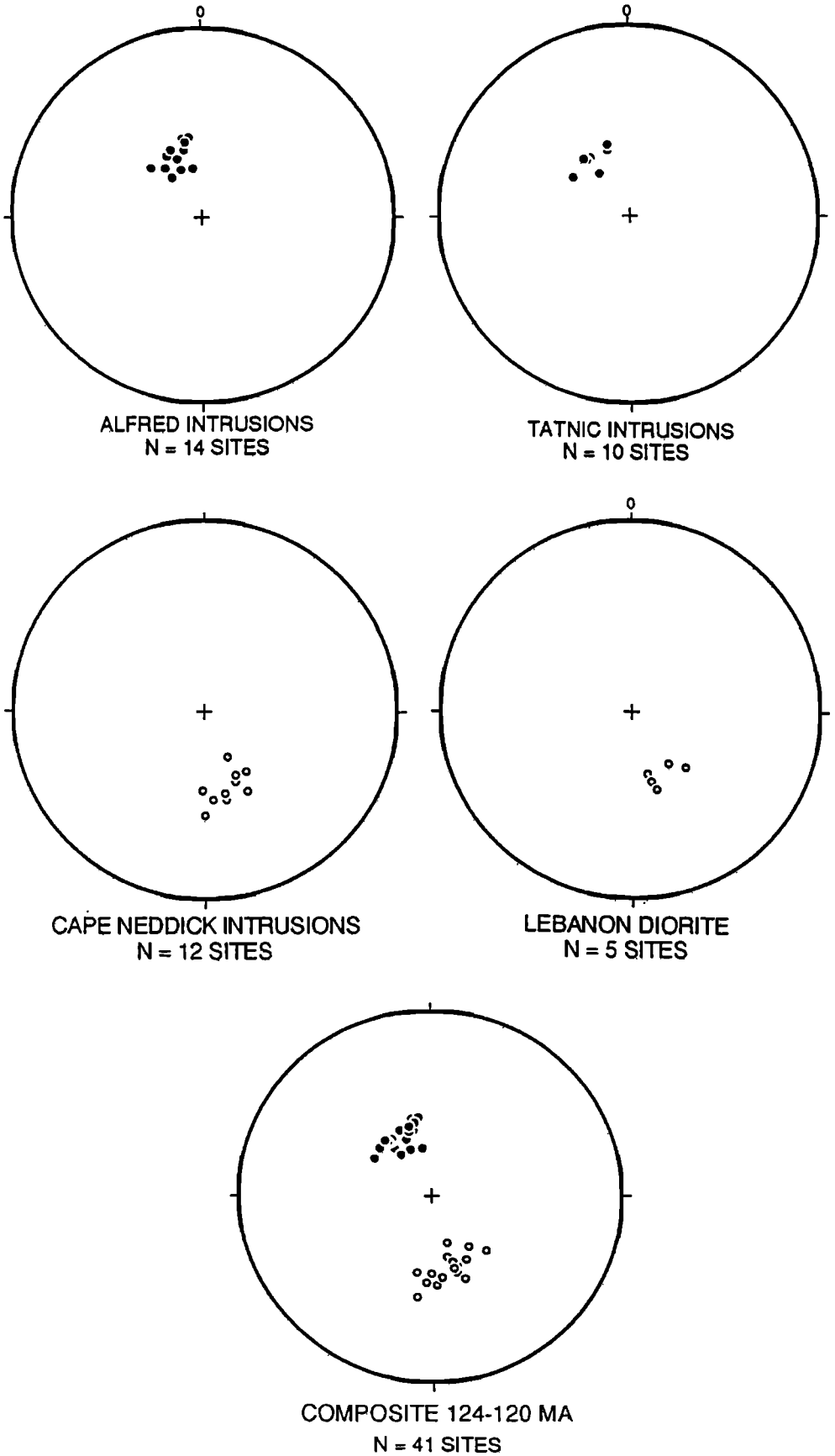


Figure 11. Equal area diagrams of site mean paleomagnetic directions from the Alfred, Tatnic, Cape Neddick, Lebanon Intrusions and composite diagram is shown. Open circles indicate upper hemisphere projections; solid circles indicate lower hemisphere projections; N is number of sites.

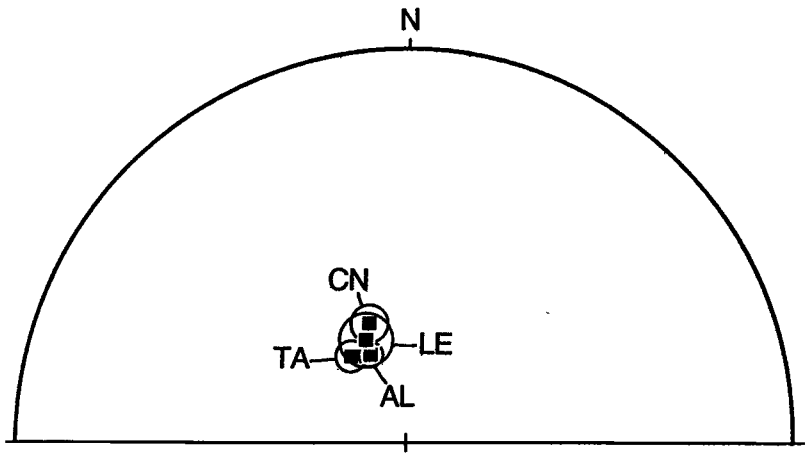


Figure 12. Equal angle diagram of mean paleomagnetic directions from the southern Maine intrusions is shown. The reversed mean site directions for the Cape Neddick and Lebanon intrusions have been inverted and plotted in the lower hemisphere to inspect for the reversal test. The mean directions overlap and are indistinguishable at the 95% confidence level.

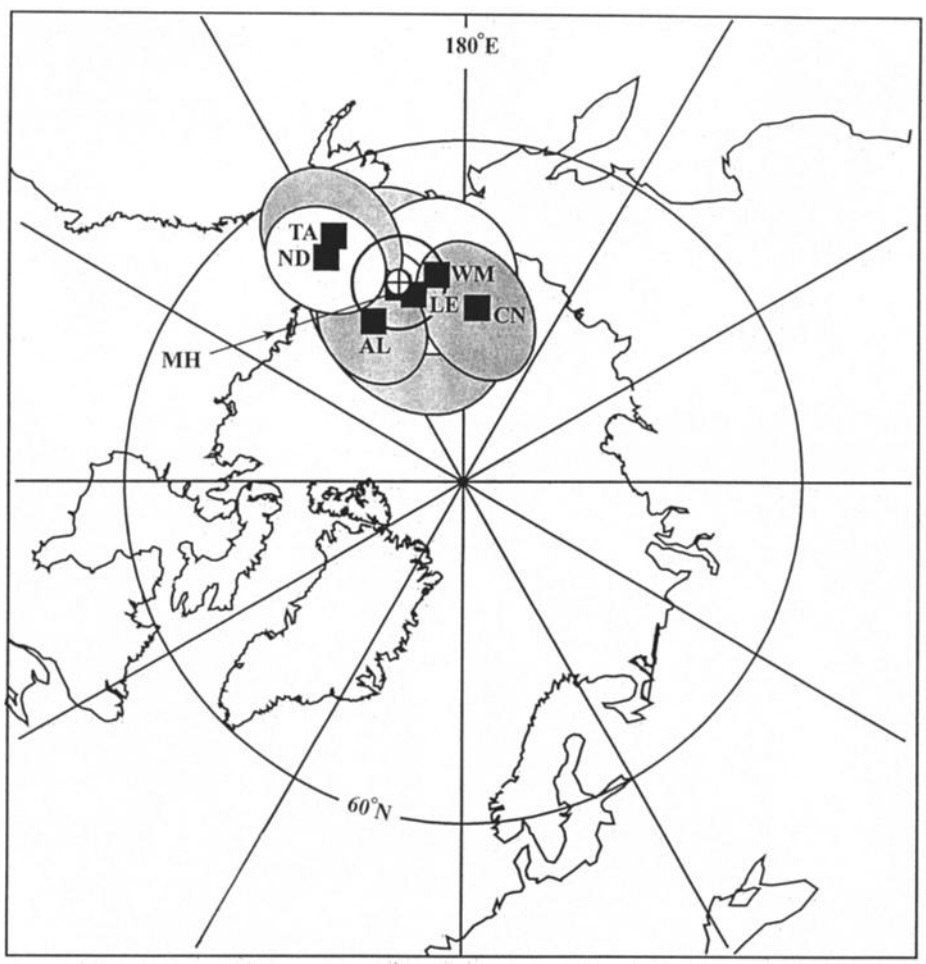


Figure 13. Early Cretaceous paleomagnetic poles (black squares) with shaded α_{95} for the Alfred (AL), Tatnic (TA), Cape Neddick (CN), and Lebanon intrusions (LE). Also plotted in black squares are other coeval poles; (ND) Notre Dame Bay dikes, Newfoundland [*LaPointe,* 1979] with unshaded α_{95}; (MH) Monteregian Hills [*Foster and Symons,* 1979] with unshaded α_{95}; and (WM) White Mountains, [*Van Fossen and Kent,* 1992] with unshaded A95. A combined Early Cretaceous paleopole (circle with cross) is shown with unshaded A95 .

Jurassic paleomagnetic poles argue against any significant post-Cretaceous tilting.

All sites from the four intrusions are plotted in Figure 11. These intrusions show excellent agreement within each polarity. After inversion of the mean directions of the reversed intrusions through the origin, the confidence limits surrounding the mean direction (α_{95}) of the two polarity groups intersect (Figure 12); the mean normal and reversed sites are not significantly different at the 95% level [*McFadden and Lowes*, 1981]. The angular dispersion about the site mean VGPs (N = 41) is low with an S = 8.8° with upper and lower limits at the 95% confidence level of S_u = 9.4° and S_l = 8.3° using the method of *Cox* [1969]. These values are lower than the predicted value (~15) for a calculated paleolatitude of ~40° for the southern Maine intrusions. It appears that paleosecular variation has been averaged in the intrusions, and it is interesting to note that all other coeval poles, the Monteregian Hills [*Foster and Symons*, 1979]; White Mountains [*Van Fossen and Kent*, 1992]; and the Notre Dame Bay Newfoundland dikes [*LaPointe*, 1979], also display lower than expected angular dispersion.

Paleopoles and Comparisons with Other Cretaceous Data

The need for high-precision paleomagnetic reference poles has increased as methods and modeling techniques have developed. Differences of only a few degrees can be crucial to plate motion studies. The cusps present in recent Mesozoic apparent pole wander (APW) paths [*May and Butler*, 1986] are the result of new modeling techniques and increased precision in paleomagnetic poles. Older techniques [*Irving and Irving*, 1982] using the sliding window method averaged out these sharp changes in plate motion. Synthetic models [*Gordon et al.*, 1984] constrain the path to small circles. High-precision data are needed to determine if plates are constrained to small circle rotations and, if so, the length of the segment of each small circle track.

The calculated paleopoles are Lebanon Diorite (125 Ma ± 3) 73.2°N, 195.1°E, α_{95} =7.5; Tatnic intrusion (122 Ma ± 2) at 65.7°N, 208.4°E, α_{95} =4.6 ; Alfred intrusion (120.4 Ma ± 1.0) at 74.0°N, 209.8°E, α_{95}=3.6 ; and 74.9°N 175.2°E, α_{95} =4.6 for the Cape Neddick Complex (120.6 Ma ± 0.4). A high-precision paleopole was calculated from the dual polarity intrusions (Table 1b.). This 125-120 Ma group paleopole is calculated to be 72.3°N, 198.9°E, A_{95} = 3.3°. The pole positions of the southern Maine intrusions are shown on Figure 13.

Three earlier studies exist for this time period. The 124±1 Ma Monteregian Hills, Quebec [*Foland et al.*, 1986), give a pole at 73°N, 191°E, α_{95} = 2.0° [*Foster and Symons*, 1979]. The 129 Ma Notre Dame Bay Newfoundland dikes give a pole at 71°N, 207°E, α_{95} = 4.0° [*LaPointe*, 1979]. The White Mountains, with an age span of 122.5-112± 5 Ma, give a mean pole of 71.9°N, 187.4°E, A_{95} = 6.9° [*Van Fossen and Kent*, 1992]. The coherence of the precisely dated southern Maine intrusions and the Monteregian Hills is striking and yields a combined pole at 72.6°N, 196.7°E, A_{95} = 4.4. When the less well age-constrained Notre Dame Bay Newfoundland dikes and the White Mountains poles are added, this produces a combined pole at 71.7°N, 198.4°E, A_{95} = 4.1 (Figure 13). In addition, the coherence in pole positions of the southern Maine intrusions, the Monteregian Hills, the Notre Dame Bay Newfoundland dikes, and the White Mountains poles argue against any significant post-Cretaceous tilting in southern Maine. Though these paleopoles are displaced northward from *May and Butler's* [1986] Jurassic-Cretaceous track, their positions are lower in latitude than the Late Cretaceous paleopoles from Granite Mountain, Arkansas [*Globerman and Irving*, 1988], Elkhorn Mountains, Montana [*Diehl*, 1991], the Adel Mountains, Montana [*Gunderson and Sheriff*, 1991], and Gringo Gulch Volcanics, Arizona [*Barnes and Butler*, 1980]. As has been suggested by many authors, there appears to have been a decrease in North American plate motion during the Early Cretaceous and a virtual standstill until the Middle to Late Cretaceous when plate motion increased. This new paleopole derived from the southern Maine intrusions would support a gradual increase in northwestward plate motion from Early to Late Cretaceous.

The Cretaceous average determined by *Mankinen* [1978], which was used by *May and Butler* [1986], and the synthetic pole calculated by *Gordon et al.* [1984], which used older data with poorer Q factors [*Van der Voo*, 1993] than recently determined poles, should not be used without elimination of the poorly determined poles from the averages and the addition of recently determined high-precision poles. The importance of a reliable Cretaceous reference pole for North America is crucial to those reconstructing the paths and rotations of displaced terranes of the Western Cordillera of North America. The study of microplate tectonics in the Cordillera has been seriously hampered by the limited North American paleomagnetic database for the Middle Jurassic and Cretaceous. The data set presented here should aid in such studies.

The well-characterized reversed polarity recorded in the precisely dated Cape Neddick Complex may represent the best date for the magnetic polarity chron M0r. This is the last reversal before the onset of the Cretaceous normal superchron. In addition, the M0r has been suggested as the Barremian-Aptian stage boundary [*Gradstein et al.*, 1994].

Summary

The Early Cretaceous paleopole reported in this account is from well-dated (125-120 Ma) intrusions in southern Maine. The Alfred and Tatnic Complexes are of normal polarity, whereas the Lebanon and Cape Neddick intrusions are of reversed polarity. A reversal test is positive at the 95% confidence level. High coercivity and high unblocking temperature magnetizations are well isolated from 40 of the 41 sites. The primary nature of the remanence carrier has been verified by detailed paleomagnetic and rock magnetic experiments, and by optical observations. It is supported by the fact that both normal and reversed directions are obtained. The dominant remanence carrier is titanomagnetite which crystallized from the magma or was produced by deuteric oxidation during post-magmatic cooling. The calculated 125-120 Ma paleopole at 72.3°N, 198.9°E (A_{95} = 3.3°) plots at higher latitude than the predicted pole for North America [*May and Butler*, 1986; *Gordon et al.*, 1984] but is in excellent agreement with the coeval and dual polarity Monteregian Hills pole of *Foster and Symons* [1979]. The Cape Neddick intrusion at 120.6 Ma ± 0.4 may represent the best date for the reversed magnetic polarity chron M0r.

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